

Substituting complex behaviour descriptions via Hardware-in-the-Loop integration with SysML system models

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Abstract

This paper describes an approach used to integrate a 3rd party subsystem into a high level MBSE description of a reactive behavioural system architecture. The reactive behavioural architecture is defined using Systems Modelling Language (SysML) within the CATIA Magic tool-set. The physical hardware behaviour was not replicated as part of the MBSE model but extracted through changes in measured values/parameters of a subsystem over time in response to inputs from simulated behavioural models of connecting subsystems. This approach enabled simulation of a subsystem which has complex interactions which are unfeasible to capture in a model in its entirety. Bidirectional communication was established to complete a feedback loop and the overall simulation was evaluated using a test harness to determine conformance to key requirements. The state of the hardware was measured through internal sensors which was passed back to a test harness established in the model for evaluation.

For testing, A prototype, scale lift system of an UAV was used as the subsystem for integration into the behavioural architecture that interacts with a simulated control system. An abstracted behavioural model of the subsystem was initially defined to provide the context for the design and integration of an appropriate control system that would be controlling a UAV. As the subsystem has a fluid dynamic interaction with the outside environment, this would be infeasible to attempt to model using SysML and was therefore deemed appropriately complex for the purposes of this paper. This paper tested the feasibility of substituting existing hardware for a detailed behavioural model description and successfully integrating with adjoining purely modelled subsystems.

This paper presents the considerations and requirements that were necessary to enable integration of hardware into a higher level of abstraction model. The paper describes the overall simulation process as well as limitations and use cases for this approach.